Curvature

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Motivation: Local Invariants

An important question about Riemannian manifolds is whether all Riemannian metrics are locally isometric.

The round 2-sphere and the Euclidean plane are not locally isometric.

We seek a local invariant that detects this difference.

Motivation: Extending a Tangent Vector

Let $p \in \mathbb{R}^2$, and let $Z_p = Z_1 \frac{\partial}{\partial x^1} \mid_p + Z_2 \frac{\partial}{\partial x^2} \mid_p \in T_p\mathbb{R}^2$.

$$Z(q) := Z_1 \frac{\partial}{\partial x^1} \mid_q + Z_2 \frac{\partial}{\partial x^2} \mid_q, \quad q \in \mathcal{M}$$

Then $\nabla_{\partial_1} Z = \nabla_{\partial_2} Z = 0$ and therefore $\nabla_{\dot{\gamma}(t)} Z \equiv 0$ for every curve γ.

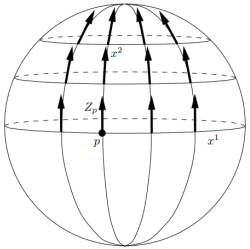
Every tangent vector can be extended to a parallel vector field.

Manifold: Extending a Tangent Vector

Let M be a Riemannian 2-manifold, let $p \in M$, and let $Z_p \in T_pM$. Choose local coordinates (x^1, x^2) centered at p.

Construct a local vector field Z in two steps.

1. Parallel translate Z_p along the x^1-axis.
2. From every point on that axis, parallel translate along the coordinate line parallel to the x^2-axis.



Manifold: Commuting Second Covariant Derivatives

Considering $\gamma(t) = (t, 0), \dot{\gamma}(t) = \partial_1 \mid_{\gamma(t)}$ and $V(t) := P_{t,t_0} Z_p$ then $0 = D_t V \mid_t = \nabla_{\dot{\gamma}(t)} \tilde{Z} \mid_{\gamma(t)}$. Therefore,

$$\nabla_{\partial_1} Z = 0 \qquad \text{when } x^2 = 0,$$

and

$$\nabla_{\partial_2} Z = 0 \qquad \text{everywhere.}$$

If $\nabla_{\partial_2} \nabla_{\partial_1} Z = \nabla_{\partial_1} \nabla_{\partial_2} Z$ holds, then $\nabla_{\partial_2}(\nabla_{\partial_1} Z) = 0$ and $\nabla_{\partial_1} Z = 0$ hold by uniqueness of parallel translation.
Thus $\nabla_{c_1 \partial_1 + c_2 \partial_2} Z \equiv 0$ and the extension succeeds precisely when these two second covariant derivatives commute.

The Euclidean Calculation

On $\mathbb{R}^n$ with the **Euclidean connection**, write $Z = Z^k \partial_k$. Then

$$\begin{aligned}
\nabla_{\partial_2} \nabla_{\partial_1} Z &= \nabla_{\partial_2} \big(\partial_1 Z^k \partial_k \big) + \nabla_{\partial_2} \big(Z^k \nabla_{\partial_1} \partial_k \big) \\
&= \partial_2 (\partial_1 Z^k) \partial_k \\
&= \partial_1 (\partial_2 Z^k) \partial_k \\
&= \nabla_{\partial_1} \nabla_{\partial_2} Z.
\end{aligned}$$

Ordinary second partial derivatives commute.

For a general Riemannian metric, this equality can fail, and that failure prevents the construction of a parallel extension.

The Flatness Criterion

For arbitrary vector fields X, Y, Z on Euclidean space,

$$\begin{aligned}\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z &= \big(XYZ^k - YXZ^k\big)\partial_k \\ &= [X,Y]Z^k \partial_k \\ &= \nabla_{[X,Y]} Z.\end{aligned}$$

Therefore Euclidean connection satisfies

$$\nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z = \nabla_{[X,Y]} Z. \tag{7.1}$$

We call (7.1) the **flatness criterion**.

The Curvature Endomorphism

Definition

The **Riemann curvature endomorphism** is the map

$$R : \mathcal{T}(M) \times \mathcal{T}(M) \times \mathcal{T}(M) \longrightarrow \mathcal{T}(M)$$

defined by

$$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z.$$

The first two terms compare the two orders of covariant differentiation.

The bracket term removes the noncommutativity already caused by the vector fields X and Y themselves.(e.g. $X = \partial_x, Y = x\partial_y$)

The Curvature Endomorphism Is a Tensor

Proposition 7.1

The curvature endomorphism is a $\binom{3}{1}$-tensor field.

Equivalently, for every $p \in M$, the value

$$R(X, Y)Z\big|_p$$

depends only on X_p, Y_p, and Z_p, and the dependence on these three vectors is multilinear.

The Curvature Endomorphism Is a Tensor

Proof.
By the tensor characterization lemma, it is enough to prove multilinearity over $C^\infty(M)$.
For $f \in C^\infty(M)$,

$$\begin{aligned} R(X, fY)Z &= \nabla_X \nabla_{fY} Z - \nabla_{fY} \nabla_X Z - \nabla_{[X,fY]} Z \\ &= \nabla_X (f \nabla_Y Z) - f \nabla_Y \nabla_X Z - \nabla_{f[X,Y]+(Xf)Y} Z \\ &= (Xf) \nabla_Y Z + f \nabla_X \nabla_Y Z - f \nabla_Y \nabla_X Z \\ &\quad - f \nabla_{[X,Y]} Z - (Xf) \nabla_Y Z \\ &= f R(X, Y) Z. \end{aligned}$$

The Curvature Endomorphism Is a Tensor

Proof (continued).
The definition gives

$$R(X,Y)Z = -R(Y,X)Z,$$

so $C^\infty(M)$-linearity in X follows from linearity in Y.

For the final argument,

$$\nabla_X \nabla_Y (fZ) = X(Yf)Z + (Yf)\nabla_X Z + (Xf)\nabla_Y Z + f\nabla_X \nabla_Y Z,$$
$$\nabla_Y \nabla_X (fZ) = Y(Xf)Z + (Xf)\nabla_Y Z + (Yf)\nabla_X Z + f\nabla_Y \nabla_X Z.$$

The Curvature Endomorphism Is a Tensor

Proof (continued).
Also,

$$\nabla_{[X,Y]}(fZ) = ([X,Y]f)Z + f\nabla_{[X,Y]}Z.$$

The derivative terms multiplying $\nabla_X Z$ and $\nabla_Y Z$ cancel.
The remaining scalar term is zero because

$$X(Yf) - Y(Xf) - [X,Y]f = 0.$$

Therefore

$$R(X,Y)(fZ) = fR(X,Y)Z.$$

Thus R is multilinear over $C^\infty(M)$. □

The Curvature Tensor and Its Components

With local coordinate (x^i), we denote $R(\partial_i, \partial_j)\partial_k = R_{ijk}{}^l \partial_l$ and therefore

$$R = R_{ijk}{}^l \, dx^i \otimes dx^j \otimes dx^k \otimes \partial_l.$$

Definition

Lowering the last index gives the **Riemann curvature tensor**

$$\mathrm{Rm}(X, Y, Z, W) = \langle R(X, Y)Z, W \rangle, \quad X, Y, Z, W \in \mathcal{T}(\mathcal{M})$$

$$\mathrm{Rm} = R_{ijkl} \, dx^i \otimes dx^j \otimes dx^k \otimes dx^l.$$

Its components satisfy

$$R_{ijkl} = \langle R(\partial_i, \partial_j)\partial_k, \partial_l \rangle = g_{lm} R_{ijk}{}^m.$$

Isometry Invariants

Lemma 7.2

If

$$\varphi : (M, g) \longrightarrow (\widetilde{M}, \widetilde{g})$$

is a local isometry, then

$$\widetilde{R}(\varphi_* X, \varphi_* Y)\varphi_* Z = \varphi_*\big(R(X, Y)Z\big),$$

and

$$\varphi^* \widetilde{\mathrm{Rm}} = \mathrm{Rm}\,.$$

Isometry Invariants

Proof.

Naturality of the Riemannian connection and the Lie bracket gives

$$\widetilde{\nabla}_{\varphi_* X}(\varphi_* Y) = \varphi_*(\nabla_X Y), \qquad [\varphi_* X, \varphi_* Y] = \varphi_*[X, Y].$$

Applying these identities to the definition of curvature yields

$$\begin{aligned}\widetilde{R}(\varphi_* X, \varphi_* Y)\varphi_* Z &= \widetilde{\nabla}_{\varphi_* X}\widetilde{\nabla}_{\varphi_* Y}\varphi_* Z - \widetilde{\nabla}_{\varphi_* Y}\widetilde{\nabla}_{\varphi_* X}\varphi_* Z - \widetilde{\nabla}_{[\varphi_* X, \varphi_* Y]}\varphi_* Z \\ &= \varphi_*\big(R(X, Y)Z\big).\end{aligned}$$

Because φ_* preserves the metric,

$$\begin{aligned}\varphi^*\widetilde{\mathrm{Rm}}(X, Y, Z, W) &= \langle \tilde{R}(\varphi_* X, \varphi_* Y)\varphi_* Z, \varphi_* W\rangle_{\tilde{g}} \\ &= \langle \varphi_*\big(R(X, Y)Z\big), \varphi_* W\rangle_{\tilde{g}} \\ &= \langle R(X, Y)Z, W\rangle_g = \mathrm{Rm}(X, Y, Z, W)\end{aligned}$$

□

Flatness Theorem

Definition

A Riemannian manifold is **flat** if every point has a neighborhood isometric to an open subset of Euclidean space.

Theorem 7.3

A Riemannian manifold is flat if and only if its curvature tensor vanishes identically.

Flatness Theorem

Proof.
First suppose that (M, g) is flat.
The Euclidean metric satisfies the flatness criterion
$R(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]} Z = 0$ so its curvature endomorphism and curvature tensor vanish identically.

Near every $p \in M$, there is an isometry

$$\varphi : (U, g) \longrightarrow (V, \overline{g}), \qquad V \subset \mathbb{R}^n.$$

By Lemma 7.2,

$$\mathrm{Rm}_g = \varphi^* \, \mathrm{Rm}_{\overline{g}} = 0$$

on U.
Thus flatness implies vanishing curvature.

Flatness Theorem

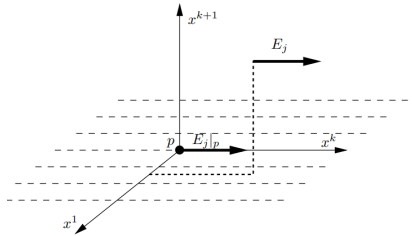
Proof (continued).

Now suppose that $\mathrm{Rm} \equiv 0$.

Then $R \equiv 0$, so the flatness criterion holds for all vector fields.

Fix $p \in M$ and choose an orthonormal basis $(E_1|_p, \dots, E_n|_p)$ of T_pM.
Choose coordinates $(x^1, \dots, x^n)$ centered at p such that $E_i|_p = \partial_i|_p$, and shrink the coordinate neighborhood to a cube.

Construct each E_j by successively translating along the x^1-, x^2-, …, x^n-coordinate lines.
Parallel translation preserves inner products, so $(E_1, \dots, E_n)$ is an orthonormal frame.



Flatness Theorem

Proof (continued).

Let

$$M_k = \{x^{k+1} = \cdots = x^n = 0\}.$$

We prove by induction on k that

$$\nabla_{\partial_1} E_j = \cdots = \nabla_{\partial_k} E_j = 0 \qquad \text{on } M_k. \tag{7.2}$$

For $k = 1$, this is true by construction.

For $k = n$, it says that E_j is parallel throughout the coordinate cube.

Assume that (7.2) holds on M_k.

Flatness Theorem

Proof (continued).
On M_{k+1}, construction gives $\nabla_{\partial_{k+1}} E_j = 0$.
For $i \le k$, the inductive hypothesis gives $\nabla_{\partial_i} E_j = 0$ on $M_k = \{x^{k+1} = \cdots = x^n = 0\}$.
Since $R = 0$ and $[\partial_{k+1}, \partial_i] = 0$,

$$\begin{aligned} \nabla_{\partial_{k+1}}\big(\nabla_{\partial_i} E_j\big) &= \nabla_{\partial_i}\big(\nabla_{\partial_{k+1}} E_j\big) \\ &= 0. \end{aligned}$$

Uniqueness of parallel translation therefore implies

$$\nabla_{\partial_i} E_j = 0 \qquad \text{on } M_{k+1}.$$

This completes the induction and proves that the orthonormal frame is parallel.

Flatness Theorem

Proof (continued).
Because the Riemannian connection is symmetric,

$$[E_i, E_j] = \nabla_{E_i} E_j - \nabla_{E_j} E_i = 0.$$

Thus $(E_1, \ldots, E_n)$ is a commuting independent frame.
The normal form theorem for commuting vector fields gives local coordinates $(y^1, \ldots, y^n)$ such that

$$E_i = \frac{\partial}{\partial y^i}.$$

In these coordinates,

$$g_{ij} = g(E_i, E_j) = \delta_{ij}.$$

Therefore y is a local isometry to Euclidean space, and M is flat. □

Ricci and Scalar Curvatures

The **Ricci curvature** is the covariant 2-tensor obtained by tracing the first and last indices of the curvature endomorphism.
Its components are

$$R_{ij} := R_{kij}{}^{k} = g^{km} R_{kijm}.$$

The **scalar curvature** is the trace of the Ricci tensor:

$$S := \operatorname{tr}_g \mathrm{Rc} = R_i{}^{i} = g^{ij} R_{ij}.$$

Thus

$$\mathrm{Rm} \longrightarrow \mathrm{Rc} \longrightarrow S$$

successively compresses curvature information.

Curvature in Coordinates

Let M be a Riemannian manifold and let (x^i) be arbitrary local coordinates.
Recall $R = R_{ijk}{}^{l}\, dx^i \otimes dx^j \otimes dx^k \otimes \partial_l$ and $\mathrm{Rm} = R_{ijkl}\, dx^i \otimes dx^j \otimes dx^k \otimes dx^l$.

Problem 7-1

(a) Compute the components of the Riemann curvature tensor in terms of the Christoffel symbols.

(b) If (x^i) are normal coordinates centered at $p \in M$, show that at p,

$$R_{ijkl} = \frac{1}{2}\left(\partial_j\partial_l g_{ik} + \partial_i\partial_k g_{jl} - \partial_i\partial_l g_{jk} - \partial_j\partial_k g_{il}\right).$$

Coordinate Formula of Riemannian curvature tensor

Since $[\partial_i, \partial_j] = 0$ and $\nabla_{\partial_i}\partial_j = \Gamma_{ij}^l \partial_l$,

$$\begin{aligned} R(\partial_i, \partial_j)\partial_k &= \nabla_{\partial_i}(\Gamma_{jk}^m \partial_m) - \nabla_{\partial_j}(\Gamma_{ik}^m \partial_m) \\ &= \big(\partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{jk}^m \Gamma_{im}^l - \Gamma_{ik}^m \Gamma_{jm}^l\big)\partial_l. \end{aligned}$$

Therefore,

$$R_{ijk}{}^l = \partial_i \Gamma_{jk}^l - \partial_j \Gamma_{ik}^l + \Gamma_{jk}^m \Gamma_{im}^l - \Gamma_{ik}^m \Gamma_{jm}^l.$$

Coordinate Formula of Riemannian curvature tensor

Let (x^i) be normal coordinates centered at p. Then

$$g_{ij}(p) = \delta_{ij}, \qquad \partial_k g_{ij}(p) = 0, \qquad \Gamma^l_{ij}(p) = 0.$$

Since $R_{ijkl} = \langle R(\partial_i, \partial_j)\partial_k, \partial_l \rangle = \langle R_{ijk}{}^r \partial_r, \partial_l \rangle$,

$$R_{ijkl}(p) = \partial_i \Gamma^l_{jk}(p) - \partial_j \Gamma^l_{ik}(p).$$

Using

$$\Gamma^l_{jk} = \frac{1}{2} g^{lm} \big(\partial_j g_{km} + \partial_k g_{jm} - \partial_m g_{jk} \big),$$

and $\partial_i g^{lm}(p) = 0$, we obtain

$$\partial_i \Gamma^l_{jk}(p) = \frac{1}{2} \big(\partial_i \partial_j g_{kl} + \partial_i \partial_k g_{jl} - \partial_i \partial_l g_{jk} \big)_p.$$

Curvature Is Second-Order Data

Therefore,

$$R_{ijkl}(p) = \frac{1}{2}\big(\partial_i\partial_j g_{kl} + \partial_i\partial_k g_{jl} - \partial_i\partial_l g_{jk} \\ - \partial_j\partial_i g_{kl} - \partial_j\partial_k g_{il} + \partial_j\partial_l g_{ik}\big)_p.$$

The first and fourth terms cancel because mixed partial derivatives commute. Thus,

$$R_{ijkl}(p) = \frac{1}{2}\left(\partial_j\partial_l g_{ik} + \partial_i\partial_k g_{jl} - \partial_i\partial_l g_{jk} - \partial_j\partial_k g_{il}\right)_p.$$

In normal coordinates, the metric and all its first derivatives look Euclidean at p.

Curvature is the remaining second-order deviation from the Euclidean metric.

Summary

- Curvature measures the corrected failure of second covariant derivatives to commute.
- $\mathrm{Rm} \equiv 0$ is equivalent to local Euclidean geometry.
- Ricci and scalar curvature are contractions of the full curvature tensor.
- In normal coordinates, curvature is exactly second-order metric data.